

# Anshul Agrawal

☎ +1 (416) 606-8534 | ✉ aanshul22@gmail.com | 📱 aanshul22 | 🌐 aanshul22

## Skills

|                    |                                                                                 |
|--------------------|---------------------------------------------------------------------------------|
| <b>Languages</b>   | Python, Rust, Go, C++, C, Java, SQL, Bash, TypeScript                           |
| <b>AI/ML</b>       | LangGraph, LangChain, MCP, Codex, Deep Agents, PyTorch, MLflow, OpenCV          |
| <b>Frameworks</b>  | Django, FastAPI, Pydantic, Celery, Redis, RabbitMQ, GraphQL, NumPy, Pandas      |
| <b>Infra/Tools</b> | Git, Docker, Kubernetes, Terraform, Helm, Jenkins, Pytest, OpenTelemetry, Nginx |
| <b>Services</b>    | Azure, AWS, Stripe, Auth0, Vault, MySQL                                         |

## Work Experience

### Layer 6 AI, TD Bank

Toronto, ON

MACHINE LEARNING ENGINEER II | PYTHON, LANGGRAPH, MCP, CODEX, DEEP AGENTS

Aug. 2025 - Present

- Built reusable frameworks for production-grade **agentic AI systems** in regulated banking, spanning document intelligence, coding agents, PR review automation, eval harnesses, tracing, and human-in-the-loop workflows.
- Ideated, architected, and developed a patent-pending **document intelligence/OCR framework** standardized as TD's central document AI framework, deployed across **8 production use cases**.
- Accelerated OCR-based delivery by **8--12 weeks** per project through reusable extraction, parsing, validation, and customization primitives; matched top vendor accuracy at **60% lower cost**.
- Designed a patent-pending **OpenTelemetry-based tracing layer** with PII/PCI redaction, file handling, and trace compaction, reducing agent trace sizes by **96%**.
- Drove company-wide agentic development adoption across **60+ users** by building coding-agent workflows, packaged skills, hooks, MCP servers, and a PR reviewer that improved merge time by **21%**.

### Lorica Cybersecurity

Toronto, ON

SOFTWARE ENGINEER | DJANGO, C++, AUTH0, STRIPE, TERRAFORM, HELM, AZURE, CELERY, RABBITMQ, NGINX

Jul. 2023 - May. 2025

- Engineered a **Django backend** and accelerated development by integrating **Auth0** for authentication, **Stripe** for payments, **Azure** for hosting, **HubSpot** for CRM, and **MySQL** for data storage.
- Engineered scripts for **infrastructure provisioning** and **deployment automation** using Terraform, Helm, Vault, Azure CLI---reducing developer overhead and slashing deployment times by **71%**.
- Led development of billing components---including subscription plans, usage metering, invoicing, tax handling, and analytics---optimized for real-time customer invoicing.
- Integrated proprietary encryption algorithms and workflows from internal R&D into backend services, forming a critical component of the product's intellectual property.
- Optimized API endpoints through query tuning, caching, and gateway configs---improving response times by **47%**.

### Goldman Sachs

Toronto, ON

SOFTWARE DEVELOPER INTERN | PYTHON, JAVA, GO, NGINX, ENVOY

Jun. 2022 - Aug. 2022

- Integrated **Envoy** into the API management platform using **Java and Go**, improving performance by **25%**.
- Created Envoy proxy proof-of-concepts across multiple config types and designed custom configuration patterns.
- Built a config generator that converted OpenAPI specs into Envoy configs and dynamically updated Envoy.

### University of Toronto, Dept. of Computer Science

Toronto, ON

RESEARCH ASSISTANT | PYTHON, PYTORCH

May. 2022 - Aug. 2022

- Optimized **Functional Principal Component Analysis** models via hyperparameter tuning.
- Implemented **Dynamic Conditional Correlation GARCH** models for covariance prediction and sample path simulation.
- Created data visualizations to estimate and analyze the authenticity of simulated sample paths.

TEACHING ASSISTANT | SYSTEMS PROGRAMMING · INTRO TO COMPUTER SCIENCE

Jan. - May. 2021 & 2022

- Led weekly labs/tutorials for 70 students in C systems programming and Python OOP
- Graded assignments, tests, and final exams across Systems Programming and Intro to Computer Science.

## Education

### University of Toronto

Toronto, Canada

COMPUTER SCIENCE SPECIALIST · STATISTICS & MATHEMATICS MINOR · HONOURS BSC

Sep 2019 - May 2023

- Focus in **Artificial Intelligence and Computer Vision**.
- **GPA - 3.87 / 4.00** (Dean's List Scholar)
- Relevant courses:
  - Software Design
  - Intro to Databases
  - Operating Systems
  - Neural Nets and Deep Learning
  - Algorithm Design & Analysis
  - Data Analysis I & II